

Data Memo

- An overview of your dataset

What does it include?

The dataset contains house prices and corresponding images for over 15,000 houses. Each house has one image, and there are 7 columns in total, including an “Image_id” column that links to images in a folder. The “n_citi” column is the label-encoded version of the city information.

Where and how will you be obtaining it? Include the link and source. If you plan on creating a web scraper or using an API, describe your plans.

<https://www.kaggle.com/datasets/ted8080/house-prices-and-images-socal>

About how many observations? How many predictors?

Observations: 15,473 different houses, each documented as a photo of its exterior

Predictors:

4, about n_citi, bed, bath, sqft

What types of variables will you be working with?

Categorical: n_citi

Numerical: bed, bath, sqft

Is there any missing data? About how much? Do you have an idea for how to handle it?

No missing data, as the set has already been curated

- An overview of your research question(s)

What variable(s) are you interested in predicting? What question(s) are you interested in answering?

We aim to predict house prices based on both structured data and image features. The model will attempt to capture how image aesthetics, combined with location and other features, influence pricing.

Name your response/outcome variable(s) and briefly describe it/them.

Price. We want to predict the price that the house sold for according to the dataset.

Which predictors do you think will be especially useful?

n_citi and sqft may be especially useful, along with features present in the home's corresponding image

Is the goal of your model descriptive, predictive, inferential, or a combination? Explain.

The goal is predictive. We want to train a model to estimate the house price using both structured data (e.g., location) and image features (e.g., house condition, style). A combination of predictive and descriptive insights might also be derived from analyzing the model's feature importance.

- Your proposed project timeline

When do you plan on having your data set loaded, beginning your exploratory data analysis, etc?

Provide a general timeline for the rest of the quarter.

Week 2 (Initial): Search and download dataset, clean images, and load the data.

Week 3 (Exploratory Data Analysis): Perform initial data cleaning and EDA, including inspecting the image data and structuring how we'll process it.

Week 4-5 (Modeling): Start developing the prediction model, using both structured data and image features. Explore neural networks or CNNs for image processing.

Week 6-7 (Evaluation): Validate the model and refine performance metrics such as RMSE or MAE. If time permits, fine-tune hyperparameters.

Week 7-9 (Conclusion): Prepare the final report and presentation, including visualizations of results and key findings from the project.

- Any questions or concerns

Are there any problems or difficult aspects of the project you anticipate?

N/A

Any specific questions you have for me/the instructional team?

One of the members in our Canvas Group has told us he has officially dropped the class and as a result, we're planning on only having 3 group members going forward. Will that be fine?